

NIST CSF 2.0 Incident Response Analysis: ICMP Flood Denial-of-Service

Simulated Case Study

Incident Summary

A multimedia company experienced an approximately two-hour denial-of-service (DoS) outage after a malicious actor flooded its internal network with ICMP traffic. The volume of incoming packets overwhelmed network services and prevented employees from accessing normal network resources. Investigation found that the firewall had not been properly configured to control the traffic entering the network.

The incident team contained the attack by blocking incoming ICMP traffic and temporarily taking non-critical network services offline so critical services could be restored. Afterward, the security team added ICMP rate limiting, source-IP verification, network monitoring, and IDS/IPS controls to strengthen the environment against similar attacks.

NIST CSF 2.0 Analysis — Govern, Identify, Protect

Govern. Set clear standards for firewall configuration, security logging, network changes, and incident-response responsibilities. These controls should have defined owners and should be reviewed regularly so important security settings do not remain incomplete or outdated.

Identify. Keep an accurate picture of the network services the business depends on, how those services connect to one another, and what normal traffic should look like. In this incident, the internal network was disrupted for approximately two hours by an ICMP flood, and the firewall configuration was identified as an important weakness that needed to be corrected.

Protect. Configure the firewall to rate-limit excessive ICMP traffic and allow only the network communication needed for legitimate business operations. Secure baseline configurations and appropriate network segmentation can provide additional protection by reducing unnecessary exposure and helping contain disruption when one part of the network is affected.

NIST CSF 2.0 Analysis — Detect, Respond, Recover

Detect. Use network monitoring, firewall logs, and IDS/IPS alerts to recognize unusual traffic before it causes a major outage. The team should understand what normal network activity looks like so sudden increases in ICMP traffic or other unusual patterns can be investigated quickly. Source-IP verification can also help identify suspicious traffic.

Respond. Maintain a clear DoS response procedure so the team knows how to contain malicious traffic, preserve useful logs, protect critical services, and communicate the incident to the right people. If necessary, non-critical services can be taken offline temporarily so the team can focus resources on keeping essential operations available while the attack is investigated.

Recover. Restore critical network services first and confirm that traffic and system performance have returned to stable levels before bringing non-critical services fully back online. After the incident, review what happened, confirm that the new firewall and IDS/IPS controls are working as intended, and update response procedures based on what the team learned.

Security Improvement Priorities

- **Strengthen firewall configuration.** Maintain an approved firewall baseline, document how ICMP traffic should be handled, and review firewall rules after major network changes, security incidents, and during regular maintenance.
- **Improve traffic monitoring and alerting.** Establish a normal traffic baseline and alert the security team when ICMP traffic or other network activity rises far outside expected levels. Earlier detection gives the team more time to respond before legitimate users lose access.
- **Prepare and test the response plan.** Maintain a DoS incident-response procedure that clearly explains containment steps, service priorities, evidence collection, escalation responsibilities, and how network services should be restored. The procedure should be tested periodically rather than waiting for another real attack to discover whether it works.